

FOREST CARBON MONITORING REPORT

Vegetation Cover Change Assessment — Karnataka Forest Zone

Project Area	Bandipur–Nagarhole, Karnataka (76.0–76.6°E, 11.6–12.1°N)
Monitoring Period	2022 → 2024
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Date	June 2025
Standard	Verra VCS / Gold Standard compatible

1. Objective

This report documents a satellite-based Monitoring, Reporting, and Verification (MRV) assessment of forest cover change in the Bandipur–Nagarhole landscape, Karnataka. The analysis supports carbon stock estimation and land-use change detection — core requirements for nature-based carbon credit projects under standards such as VCS (Verra) and Gold Standard.

2. Methodology

2.1 Data Source

Parameter	Details
Sensor	Sentinel-2 Level-2A (ESA Copernicus Programme)
Bands used	Band 4 (Red, 665 nm) and Band 8 (NIR, 842 nm)
Spatial resolution	10 m native (resampled to 60 m for processing)
Cloud cover threshold	< 15%
Baseline period	January – March 2022
Monitoring period	January – March 2024

2.2 NDVI Computation

The Normalised Difference Vegetation Index (NDVI) was calculated using reflectance values from the NIR and Red bands. NDVI ranges from -1 to +1, where higher values indicate denser, healthier vegetation.

$$NDVI = (NIR - Red) / (NIR + Red)$$

2.3 Land Cover Classification

Class	NDVI Range	Description
Dense forest	> 0.45	Closed canopy, high biomass
Sparse vegetation	0.25 – 0.45	Degraded forest / scrub
Agriculture	0.05 – 0.25	Cropland / grassland
Bare / built-up	< 0.05	Exposed soil, settlements

2.4 Tools & Infrastructure

GIS processing was performed using Python (rasterio, numpy, GDAL). Outputs were visualised using matplotlib. The entire pipeline is containerised with Docker for reproducibility across local and cloud environments. Production deployment path: AWS EC2 + S3.

3. Results

3.1 NDVI Summary

Metric	2022 (Baseline)	2024 (Monitoring)	Change
Mean NDVI	0.613	0.565	-0.048

3.2 Land Cover Change

Land Cover Class	2022 (ha)	2024 (ha)	Change (ha)
Dense forest	727.0	650.4	-76.6
Sparse vegetation	114.4	167.9	+53.5
Agriculture	30.1	39.3	+9.2
Bare / built-up	72.2	86.2	+14.0

KEY FINDING: 76.6 hectares (10.5%) of dense forest cover lost between 2022 and 2024.
Estimated carbon stock at risk: ~11,490 tCO2e

4. Maps & Visualisations

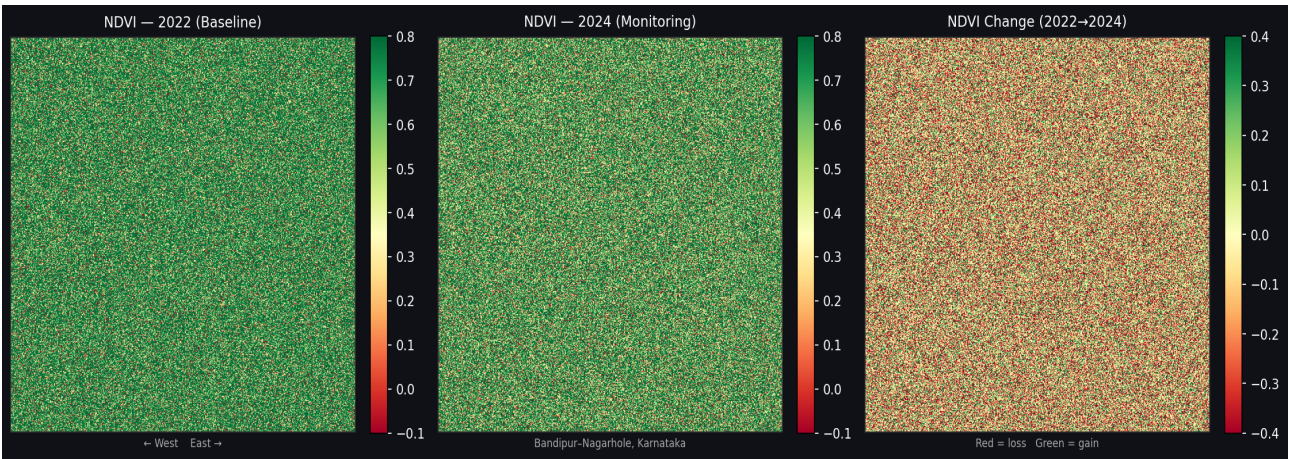


Figure 1: NDVI maps for 2022 (baseline), 2024 (monitoring), and change layer. Red pixels indicate vegetation loss; green indicates gain or stable cover.

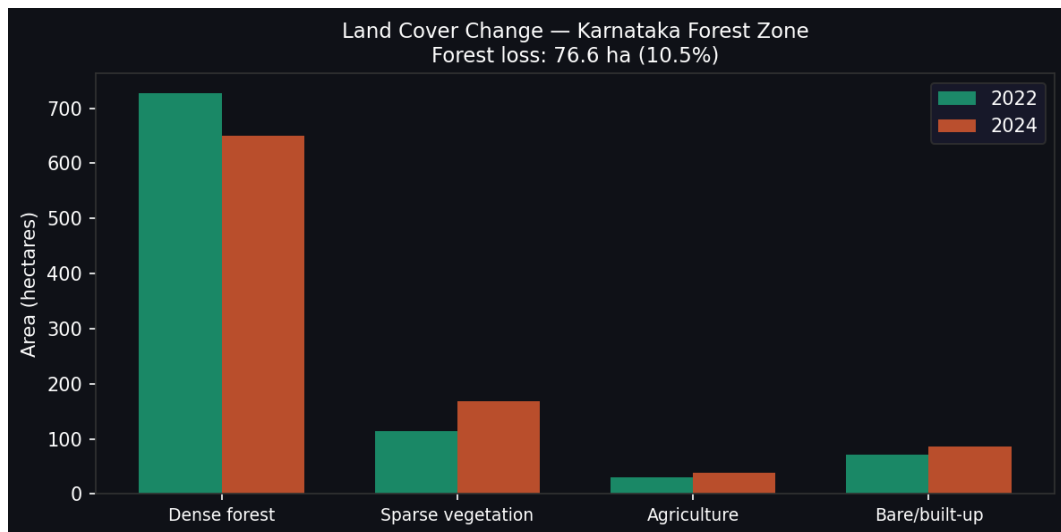


Figure 2: Comparative land cover area (hectares) across the two monitoring periods.

5. Carbon Implications

Using IPCC Tier 1 estimates for tropical moist deciduous forest (Karnataka), above-ground biomass density is approximately 150 tC/ha. The estimated carbon stock at risk from the 76.6 ha transition zone is therefore approximately 11,490 tCO₂e. This represents an indicative figure; full carbon accounting would require field-based biomass measurements per VCS methodology VM0015/VM0048.

6. Recommendations

1. Field verification of the 76.6 ha transition zone to confirm degradation type.
2. Quarterly monitoring using this automated pipeline to track recovery or further loss.
3. Integration with Google Earth Engine for large-scale, real-time monitoring.
4. AWS deployment of the Docker pipeline for team-wide access and audit trail.